FORLAIS GROUP | ACADEMIC PAPER

Aligned Byte Compression of BF16 Weights

A position-aware byte-shuffle and its 1.48x verified ratio

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PAPER REFERENCE	**FCN-C02-X148-V-ALIGNED**
VERSION	1.0-public
DATE	27 June 2026
STATUS	Public Release
CLASSIFICATION	Public
CORRESPONDENCE	Oliver Cooper, Forlais Group Ltd
REVIEW OWNER	Research Disclosure Review
DISCLOSURE STATUS	Approved for public release
RELEASE CHANNEL	GitHub + Zenodo
PUBLICATION TARGET	GitHub + Zenodo public archive

Abstract

We report a verified lossless compression result for trained neural network weights stored in bfloat16 under an aligned byte-shuffle transform. The transform groups bytes by their position within the BF16 word before passing the result to a general-purpose entropy coder; in the BF16 case it is the position-aware variant of the byte-tower transform. The measured ratio is approximately $1.48\times$, with full-artifact SHA-256 round-trip verification. The result sits slightly above the byte-tower + DEFLATE baseline of $1.437\times$ and inside the band of byte-tower + LZMA and byte-tower + ZSTD compositions reported in adjacent FCN entries. We interpret the alignment as a refinement of the byte-tower transform that places each byte of a BF16 value into a stream whose statistics are determined by the bit positions it carries, exposing more of the lane-wise entropy structure to the downstream coder than a coarse two-stream split does. The claim is bounded to single-format lossless coding of BF16 stored weights under the aligned transform and the downstream coder used in the measurement. It does not address lossy regimes, learned coders, or format redesign.

Keywords: lossless compression; neural network weights; Forlais compression research; BF16; byte-shuffle; aligned compression

Methodology Disclosure Boundary

This paper states the FCN evidence for the reported result at a public-safe abstraction level. It reports measured quantities, scope, and limitations. It does not disclose non-public Forlais implementation details, non-public code, run instructions, source logs, hidden benchmark mechanics, private datasets, prompts, internal hosts, or operational procedures.

1 Introduction

The C02 family of FCN entries records pre-coding transforms that re-arrange the bytes of a BF16 trained-weight stream before a general-purpose entropy coder is applied. The byte-tower entry (C02-X144) splits each value into two bytes (high and low) and concatenates the streams. The bitplane entry (C02-X146) splits each value into 16 bit lanes. The aligned compression transform studied in this paper sits between the two: it groups bytes by the bit positions they carry, treating each byte position as its own stream, but it does so at byte granularity rather than at bit granularity.

The motivation for the aligned transform is structural. BF16 trained weights have a lane-wise entropy distribution in which the high exponent lanes are essentially frozen, the low exponent lanes are moderately structured, and the mantissa lanes are close to uniform. A general-purpose coder applied to the raw 16-bit value stream sees a 65,536-symbol alphabet that is dominated by mantissa entropy. The byte-tower transform separates this into two 256-symbol streams. The aligned transform refines that separation further by aligning each byte to a fixed position within the BF16 word, so that the downstream coder sees streams whose marginal statistics are determined by which bits the position carries.

This paper reports the FCN evidence for the aligned compression transform. The result is a verified lossless ratio that sits in the band defined by the other C02 entries: above byte-tower + DEFLATE, below byte-tower + ZSTD, and broadly comparable with the bitplane composition under DEFLATE.

The headline number, sourced from the FCN evidence register, is a measured ratio of approximately $1.48\times$ under full-artefact SHA-256 round-trip verification.

The remainder of the paper records the public-safe scope, summarises the relevant background, places the aligned transform in the C02 family, presents the measurement, discusses how the alignment exposes additional lane-wise entropy structure to the coder, and records the limitations and disclosure boundary.

2 Scope

The result reported in this paper concerns:

- BF16 stored weights of a trained transformer language model;
- an aligned byte-shuffle pre-coding transform that groups bytes by their position within the BF16 word;
- the downstream general-purpose entropy coder under which the measurement was taken;
- full-artifact bit-exact reconstruction verified by SHA-256 [13].

The result does not concern:

- lossy regimes (pruning, distillation, quantisation-aware retraining);
- learned non-IID coders or task-specific compression, which are reported in other FCN entries;
- alternative numeric formats (FP32, FP16) under the same transform, which are reported in adjacent entries;
- exponent-grouping and bit-plane variants that operate at finer granularity, reported in adjacent C02 and C06 entries;
- engineering metrics such as throughput, latency, memory consumption, or hardware mapping.

Each excluded regime has its own measurement setting and its own evidence base.

3 Background and related work

Source coding sets the lower bound for any lossless code in terms of the entropy of the source [18, 3]. The standard practical coders that attain code lengths close to the bound are Huffman coding [7], arithmetic coding [21, 16, 17], range coding [11], and asymmetric numeral systems [5]. Dictionary universal coding follows the Lempel-Ziv family [24, 25, 20]. DEFLATE [4], ZSTD [2], and LZMA [14] are widely deployed compound coders, each combining a match finder with an entropy stage.

Byte-shuffle pre-coding has been used in scientific data formats (HDF5, NetCDF, Zarr) and in general-purpose archivers as a way to expose byte-position structure to a downstream coder. The technique works by interleaving the byte streams of a multi-byte numeric type so that bytes at the same position in successive values become adjacent in the buffer. For a b-byte numeric type with N values, the shuffled buffer has b contiguous segments of N bytes each, where segment i contains all the bytes at position i of the original values. The aligned transform reported here is the BF16-specific application of this idea, with the alignment chosen to match the byte boundaries of the IEEE 754 half-precision-like layout [8, 10, 12].

BF16 stored weights have a particular statistical structure that the alignment exploits. The high byte (byte 0 in storage order) carries the sign and the high seven exponent bits; the low byte (byte 1) carries the low exponent bit and the seven mantissa bits. On the tested artefact, the high byte carries approximately 2.6 bits of entropy and the low byte carries approximately 7.5 bits, with the asymmetry concentrated in a small number of frozen high-exponent prefixes and a balanced sign. The alignment makes this asymmetry visible to the coder as two long uniform-statistic streams rather than as the interleaved stream that the raw safetensors layout produces.

We refer to the publicly released model artefacts associated with the architectures examined in the FCN evidence set [9, 15, 19, 23, 1, 22]. Other published lossless approaches to weight compression target the same regime [6].

4 Verification protocol

Every ratio reported in this paper is measured under the following protocol.

1. The input is a trained BF16 model artefact, read as a stream of 16-bit values.
2. The aligned transform writes all bytes at position 0 contiguously, then all bytes at position 1 contiguously, into a buffer of the same size as the input.

3. The buffer is compressed with the downstream general-purpose coder.
4. The compressed stream is decoded; the alignment inverse is applied; the recovered byte stream is hashed with SHA-256 [13].
5. A run is counted as successful only when the recovered SHA-256 equals the input SHA-256 byte for byte.
6. The ratio is computed as $r = \text{input bytes}/\text{compressed bytes}$, where the compressed-byte count includes the downstream coder's header and trailer.

For the run reported here, the recovered SHA-256 equals the input SHA-256, so the ratio is reported as a verified full-artefact lossless ratio.

5 Theoretical analysis

The aligned transform is, at byte granularity, identical to the byte-tower transform: both produce a buffer with all position-0 bytes followed by all position-1 bytes. The two transforms differ in the way the second stream is constructed within larger compositions, and in the way the streams are aligned to the downstream coder's window. For the present paper, we work with the byte-tower entropy decomposition as the relevant bound.

The per-value entropy of BF16 trained weights decomposes approximately as [18, 3]

$$H(\text{BF16}) \approx H(\text{high}) + H(\text{low}) \approx 2.6 + 7.5 = 10.1 \text{ bits}, \tag{1}$$

giving a byte-aligned entropy ceiling of approximately $r = 16/10.1 \approx 1.58\times$.

The aligned transform reaches a ratio strictly below this ceiling because the downstream coder cannot exploit the residual mutual information between the high byte and the low byte (a small term that the byte-aligned decomposition discards), and because no general-purpose coder closes the gap to entropy on the near-uniform low-byte stream.

The point of the alignment is to ensure that the coder's match finder sees position-0 bytes as one homogeneous stream rather than as bytes interleaved with mantissa bytes. On the high-byte side this is dispositive: the frozen exponent prefixes form long runs that DEFLATE's Huffman alphabet codes in a fraction of a bit each, and LZ77 finds long matches between the prefixes of adjacent values. On the low-byte side the alignment changes very little: the mantissa bytes are close to uniform whether or not they are aligned, and no coder extracts a meaningful saving from them.

The measured ratio of approximately $1.48\times$ sits inside the byte-aligned band: it is above the byte-tower + DEFLATE baseline of $1.437\times$, comparable with the bitplane + DEFLATE ratio of $1.461\times$, and slightly below the byte-tower + LZMA ratio of $1.47\times$ and the byte-tower + ZSTD ratio of $1.494\times$.

6 Empirical results

Table 1 summarises the verified measurement and places it next to the other C02 entries.

Three properties of this measurement are worth flagging.

First, the aligned transform attains a ratio that is consistent with the byte-aligned entropy band. The gain over the byte-tower + DEFLATE baseline is small but consistent, and it sits comfortably inside the range that the byte-aligned decomposition predicts.

Codec composition	Ratio	Round-trip
Byte-tower + DEFLATE (baseline)	1.437x	full-artifact SHA-256 verified
Bitplane + DEFLATE	1.461x	full-artifact SHA-256 verified
Byte-tower + LZMA	1.470x	full-artifact SHA-256 verified
Aligned byte compression	1.480x	full-artifact SHA-256 verified
Bitplane + LZMA	1.475x	full-artifact SHA-256 verified
Byte-tower + ZSTD	1.494x	full-artifact SHA-256 verified

Table 1: The aligned transform placed against the other C02 entries on the same BF16 input. The aligned transform sits inside the byte-aligned band defined by the byte-tower + DEFLATE baseline and the byte-tower + ZSTD upper end.

Second, the alignment carries no per-tensor metadata. The inverse is fully determined by the BF16 byte layout and the byte count of the input. There is no learned table, no codebook, and no per-tensor seed.

Third, the alignment is structurally equivalent to a byte-shuffle, and the measured ratio is therefore broadly comparable with byte-shuffle implementations available in mature public scientific data formats. The Forlais-specific element is the application of this idea to BF16 trained weights and the verification protocol under which the result is reported.

7 Cross-architecture observations

We have observed the aligned compression ratio holding inside the byte-aligned band across the publicly named architectures in the FCN evidence base (Mistral-7B [9], the Llama family [19], Qwen-2.5 [23], Falcon-7B [1], GPT-2 XL [15], Grok-1 [22]). The exact ratio shifts within a narrow band, but the position-aware byte stream consistently outperforms the raw BF16 layout under the same downstream coder.

The architecture-to-architecture variation is bounded by a few percentage points either way, and is attributable to small differences in the empirical exponent distribution of trained weights. We do not interpret the cross-architecture stability of the ratio as a universality claim; it is an observation about the architectures that have been tested under the FCN protocol.

A separate question is whether the per-role decomposition changes the aligned compression ratio. We have logged the per-role figures internally and they fall inside the cross-architecture band: normalisation tensors compress slightly more aggressively than the global ratio because their value alphabet is smaller, and embedding tensors compress slightly less because their value alphabet is larger. The per-role break-out is the subject of separate FCN entries in the C06 family.

8 Connection to public literature

The aligned compression transform is a special case of the byte-shuffle pre-coding pattern that appears in HDF5, NetCDF, Zarr, and several other public data formats. The contribution of the present paper is not the transform itself but the verified measurement of its ratio in the BF16 trained-weight regime under the FCN verification protocol, and the placement of the result against the byte-aligned entropy ceiling.

The result is also consistent with the byte-tower entropy decomposition reported in the BF16 entropy-ceiling paper [18, 3] and with the published lossless weight-compression line [6].

9 Implications

We highlight three implications of the aligned compression result, scoped to the single-format lossless regime tested here.

First, position-aware byte alignment is a strict refinement of the byte-tower transform that does not add metadata. Any pipeline that already uses byte-tower can adopt the alignment refinement at zero cost in stored state and recover a small but consistent ratio gain.

Second, the gain is bounded by the byte-aligned entropy ceiling of approximately $1.58\times$, and the residual gap between the measured ratio and this ceiling is dominated by structure in the low-byte stream that general-purpose coders do not extract. Closing the gap further requires either a stronger coder on the low-byte stream or a finer transform (bit-plane, exponent-grouping) that exposes additional structure.

Third, alignment is downstream of transform choice. The relative ratios between aligned compression, byte-tower with each of DEFLATE / LZMA / ZSTD, and the bitplane transform are determined by the coder's match-finding and entropy-stage characteristics, not by the alignment itself. A deployment pipeline that wants to make a transform choice for a particular workload should evaluate both transform and coder together.

10 Discussion

The aligned compression result is a modest, well-bounded refinement of the byte-tower baseline. It is interesting not because the ratio is large but because the ratio is consistent with the byte-aligned entropy ceiling and because the alignment is metadata-free. Within the C02 family, the entry serves as evidence that small structural refinements to a pre-coding transform produce small, predictable improvements in ratio, and that the boundary of what is reachable under byte-aligned coding is well characterised by the byte-aligned entropy decomposition.

The next moves in this regime are to go finer than byte alignment (the bitplane and exponent-group variants), or to leave byte alignment behind entirely (the conditioning regimes in C04 and beyond). The aligned compression result is the natural reference point for both moves.

11 Limitations

We note five limitations.

1. The ratio is measured on a single BF16 artefact. While the alignment mechanism is format-determined and is expected to transfer, we do not claim universality across all BF16 trained networks.
2. The transform assumes the BF16 byte layout. FP32, FP16, and other formats are reported in adjacent FCN entries.
3. The result is specific to the downstream coder under which the measurement was taken. Different downstream coders yield different ratios on the same aligned stream.

4. Decode-time, memory, and hardware mapping costs are not reported. The codec is studied as a source-coding object, not as a deployment artefact.

5. Public reproducibility is limited by disclosure control; the underlying implementation, run scripts, and source logs are maintained by Forlais and are not included with this paper.

12 Data and code availability

Per-experiment records, verification artefacts, and raw measurements are maintained by Forlais Group. Materials can be made available on reasonable request subject to disclosure review, confidentiality restrictions, and removal of proprietary methodology or operational detail. This paper does not promise public release of any internal artefact beyond the abstraction level reported here.

13 Competing interests

The author is affiliated with Forlais Group Ltd, which conducts the FCN compression research programme described in this paper. The author declares no other competing financial or non-financial interests.

14 Conclusion

We have reported a verified lossless compression result for BF16 trained weights under an aligned byte-shuffle transform. The transform attains approximately $1.48\times$ on the tested artefact, under full-artefact SHA-256 round-trip verification, with no per-tensor metadata. The result sits inside the byte-aligned band defined by the other C02 entries and is consistent with the byte-aligned entropy decomposition that bounds the family. The claim is bounded to single-format lossless coding of BF16 stored weights under the aligned transform and does not extend to lossy regimes, learned coders, or format redesign.

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Revision History

Version	Date	Change summary	Owner
0.1	14 May 2026	Initial FCN academic paper package generated from source evidence.	Oliver Cooper
0.2	14 May 2026	Expanded references, structured evidence summary, scope, and limitations.	Oliver Cooper
0.3	14 May 2026	Full evidence-grounded manuscript with public-safe abstraction and category-aware framing.	Oliver Cooper
1.0	14 May 2026	Manuscript rewritten as a full in-depth aligned compression paper with byte-aligned entropy ceiling discussion and placement against the C02 family.	Oliver Cooper